Hospital Operations & Billing Intelligence Dashboard:

Business Requirements Document (BRD)

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1. Project Title

Hospital Operations & Billing Intelligence Dashboard

2. Project Objective

To analyse hospital admission records and build an interactive Power BI dashboard that shows billing patterns, length of stay, admission types and insurer performance, so management can find where cost and bed-days can be reduced.

3. Business Problem / Opportunity

A hospital network holds 55,500 admission records (May 2019 to May 2024) but no single view of what drives billing and length of stay. Management cannot easily tell which conditions, insurers and admission types generate the highest cost, or how admissions and billing change over time. The dashboard should support decisions on resource planning, insurer negotiations and cost control.

4. Target Audience

Primary: hospital administrators and the finance head (CFO). Secondary: operations managers and department heads.

5. Scope

Included: admission volume, billing, length of stay, medical conditions, insurers, admission types, age groups, medications, test results and trends from 2019 to 2024.

Excluded: clinical outcome prediction, individual patient tracking, hospital-level or doctor-level ranking (the data has too many unique names), real-time data.

6. Data Source(s)

Kaggle "Healthcare Dataset" (healthcare_dataset.csv), public domain (CC0). CSV format, 55,500 rows × 15 columns, about 8 MB. Admission dates run from 8 May 2019 to 7 May 2024. Known limitations: the data appears synthetic, about 534 duplicate rows, 108 negative billing amounts, inconsistent name casing, and Hospital and Doctor columns with almost one value per row.

7. Key Metrics / KPIs
(insert a table with these columns: KPI, Definition, Business meaning, Unit)

KPI	Definition	Business meaning	Unit
Total Admissions	Count of admission records	Patient volume	count
Total Billing	Sum of Billing Amount	Revenue billed	$
Avg Billing per Admission	Total Billing ÷ Total Admissions	Cost per patient	$
Avg Length of Stay	Average of (Discharge Date – Admission Date)	Bed usage	days
Emergency Admission %	Emergency admissions ÷ Total Admissions	Pressure on services	%
Abnormal Test Result %	Abnormal results ÷ Total Admissions	Case severity	%

8.Deliverables
BRD and FRD, column assessment and cleaned dataset, dashboard mockup, Power BI dashboard (.pbix) and PDF export, analysis report, README and organised GitHub repository.

9.Timeline/Milestones(inserta table: Day, Milestone)
Day 1 dataset, BRD and setup. Day 2 assessment. Day 3 cleaning. Day 4 FRD, mockup and draft model. Day 5 data model and DAX. Day 6 dashboard. Day 7 analysis and validation. Day 8 final submission.

10. Notes / Assumptions

The data appears synthetic, so insights are for demonstration. Length of Stay = Discharge Date – Admission Date. One row represents one patient admission. Negative billing values and duplicates will be investigated before any deletion. Patient names will be excluded from the dashboard and the cleaned shared file. Billing is assumed to be in US dollars (not stated in the data).